
Urban Rig Europe — Business Plan

Version 2.0 — March 2026

1. Executive Summary

Urban Rig Europe will deploy a patented, continuous superheated-steam pyrolysis system that converts mixed, unsorted municipal and industrial waste into four marketable oil fractions, biochar, and recoverable metals — with zero dioxin emissions. The company is the European operational arm of OneWorld Corporation (Osaka, Japan), which owns and manufactures the technology.

The first deployment targets France's Île-de-France region, where 6.2 million tonnes of waste are generated annually, 55% of which is incinerated or landfilled. EU Circular Economy mandates (65% recycling, max 10% landfilling by 2035) create regulatory urgency that no member state currently meets. Urban Rig addresses this gap by transforming waste management from a cost centre into a revenue-generating operation.

A single URC-2000 plant (200 T/day nominal capacity, 140 T/day actual weight, 46,200 T/year) generates €16.1M in annual revenue from five streams: gate fees (33%), pyrolysis oil (39%), char/biochar (9%), recovered metals (4%), and carbon credits (15%). At steady state, EBITDA reaches €8.9M (55% margin) on total CAPEX of €63.8M. Simple payback is approximately 7 years, with annualised ROI on equity of 17.6%.

The technology is not a concept: it has been tested at industrial scale (Asada Shokai, Noda, Japan, 2022), independently certified for oil quality (NKKK) and emissions (KANSO Technos, 5–50× below French incineration standards), and is operating in the field (Shandong Qufu, China).

Source: UR-Financial-Model-Restructured-V3.xlsx; Asada Shokai test report; KANSO Technos emissions report; NKKK certification.

2. Company Overview

2.1 Corporate Structure

Urban Rig Europe is the regional sales and programme management entity for the European market. OneWorld Corporation (Osaka, Japan) serves as R&D headquarters, technology owner, and equipment manufacturer. The two entities operate as an integrated unit: Urban Rig Europe manages market development, client relationships, regulatory compliance, and project delivery; OneWorld Corporation manufactures equipment, maintains the IP portfolio, and provides technical support.

2.2 Technology Credentials

- **Asada Shokai industrial test programme** (Noda factory, Chiba Prefecture, 2022) — independent measurement by Land Brain Co., Ltd.
- **KANSO Technos** independent exhaust gas analysis — emissions 5–50× below French incineration standards
- **NKKK** (Japan Marine Survey Association) — certified oil quality: flashpoint 46°C, sulfur <0.2%, 43.27 MJ/kg
- **Field operations** — processing plant in Shandong Qufu (China) using recovered diesel oil as operational fuel
- **Sea Waste Zero Award** — Innovation category, Japan Foundation / Ministry of the Environment (June 2019)
- **“Plastics Smart” designation** — Ministry of Economy, Trade and Industry, Japan

Source: Asada Shokai report; KANSO Technos report; NKKK certification; OneWorld Corporation records.

2.3 Mission

To replace incineration and landfilling across Europe with a profitable, zero-emission resource recovery technology, converting waste into fuels, materials, and carbon savings at industrial scale.

3. Market Opportunity

3.1 The Global Waste Problem

The World Bank estimates 2.01 billion tonnes of MSW were generated worldwide in 2016, projected to reach 3.4 billion tonnes by 2050. In Europe, Eurostat reports 225 million tonnes in 2021 (513 kg per capita), with only 49.6% recycled or composted, 27.5% incinerated, and 22.9% still landfilled. The EU Circular Economy Package mandates 65% recycling by 2035 and maximum 10% landfilling — targets no member state currently meets.

Source: World Bank “What a Waste 2.0” (2018); Eurostat Municipal Waste Statistics (2023).

3.2 France — Primary Target Market

France generates 38.5 million tonnes of MSW annually (580 kg per capita, above the EU-27 average of 505 kg). Treatment relies heavily on incineration (38.1%) and landfilling (18.1%), with recycling at only 26.9%. Approximately 21.6 million tonnes per year is either incinerated or landfilled — waste that could be redirected to resource recovery.

French tipping fees provide strong economic incentive: landfilling costs €140–180/tonne, incineration €80–120/tonne. Urban Rig’s dual revenue model (gate fees + product sales) fundamentally changes waste economics from pure cost to profit generation.

Source: ADEME 2022; Ministry of Ecological Transition.

3.3 Île-de-France — Initial Deployment Region

The Île-de-France region (12.3 million population, 18% of France) generates 6.2 million tonnes/year (17,000 tonnes/day), with 37% going to incineration and 18% to landfill. Paris-Charles de Gaulle Airport alone produces an estimated 80,000–100,000 tonnes/year of mixed waste. The region’s density, infrastructure constraints, and sustainability pressure make it an ideal first market.

Source: ORDIF (Île-de-France Regional Waste Observatory); Annex B.

3.4 Addressable Feedstock Segments

Urban Rig’s anoxic steam cracking process is feedstock-agnostic — it profitably processes waste streams that conventional recycling and incineration handle poorly or not at all.

Feedstock Segment	France Volume/yr	Oil Yield	Gate Fee	Supply Concentration
Industrial Pure Plastics	~2.2 MT	70–82%	€60–120/T	Excellent — industrial parks; consistent; contractable
MSW (residual)	~21.6 MT	20–66%	€80–180/T	Excellent — daily municipal collection; enormous volume
End-of-Life Tyres	~530,000 T	40–50%	€0–50/T	Moderate — ÎdF ~95,000 T in ~5 weeks; batch not baseload
Aircraft & Quarantine	~80–100,000 T (CDG)	25–40%	€120–200/T	Excellent at CDG — single site; captive volume
Medical / DASRI	~700,000 T total	30–50%	€800–1,500/T	Good — hospitals in urban areas; ÎdF highest density
Automotive Shredder Residue	~200–350,000 T	35–50%	€100–200/T	Good — ~40 shredder sites; large volumes per site
Composite & Multi-Layer	~500,000 T+	40–60%	€80–150/T	Moderate — via sorting centres; EPR partnerships

The optimal strategy is a blended feedstock portfolio: MSW or airport waste as baseload, supplemented by high-margin batches of industrial plastics, tyres, and medical waste.

Source: ADEME 2022; Aliapur annual report; ETRMA; Eurostat industrial waste statistics; Annex A & B.

3.5 Competitive Landscape

Conventional incineration generates dioxins, CO₂, and toxic bottom ash (machefer). Material recovery is only 5–10%. Safe incineration plants are prohibitively expensive and large-scale. Current processing relies on tipping fees alone with no resource recovery revenue, making waste management a pure cost centre.

Mechanical recycling is limited: 76% of plastics in French MSW are either not recycled or have very low recyclability. PP has a 90% non-recycling rate, PS 95%, and composite plastics 100%.

Urban Rig’s competitive advantage is threefold: it processes unsorted mixed waste (no pre-treatment required), generates multiple revenue streams from outputs (not just tipping fees), and achieves emissions 5–50× below regulatory limits without expensive filtration systems.

Source: Annex A — Current Status Analysis.

4. Technology

4.1 Process Overview

Urban Rig is a patented superheated-steam pyrolysis and carbonization system. Operating continuously at 600°C in a completely anoxic (oxygen-free) reactor environment, it converts mixed, unsorted waste into four marketable oil fractions, industrial-grade charcoal (carbide), and recoverable metals and glass.

Overheat temperature of 7-continuous process unit

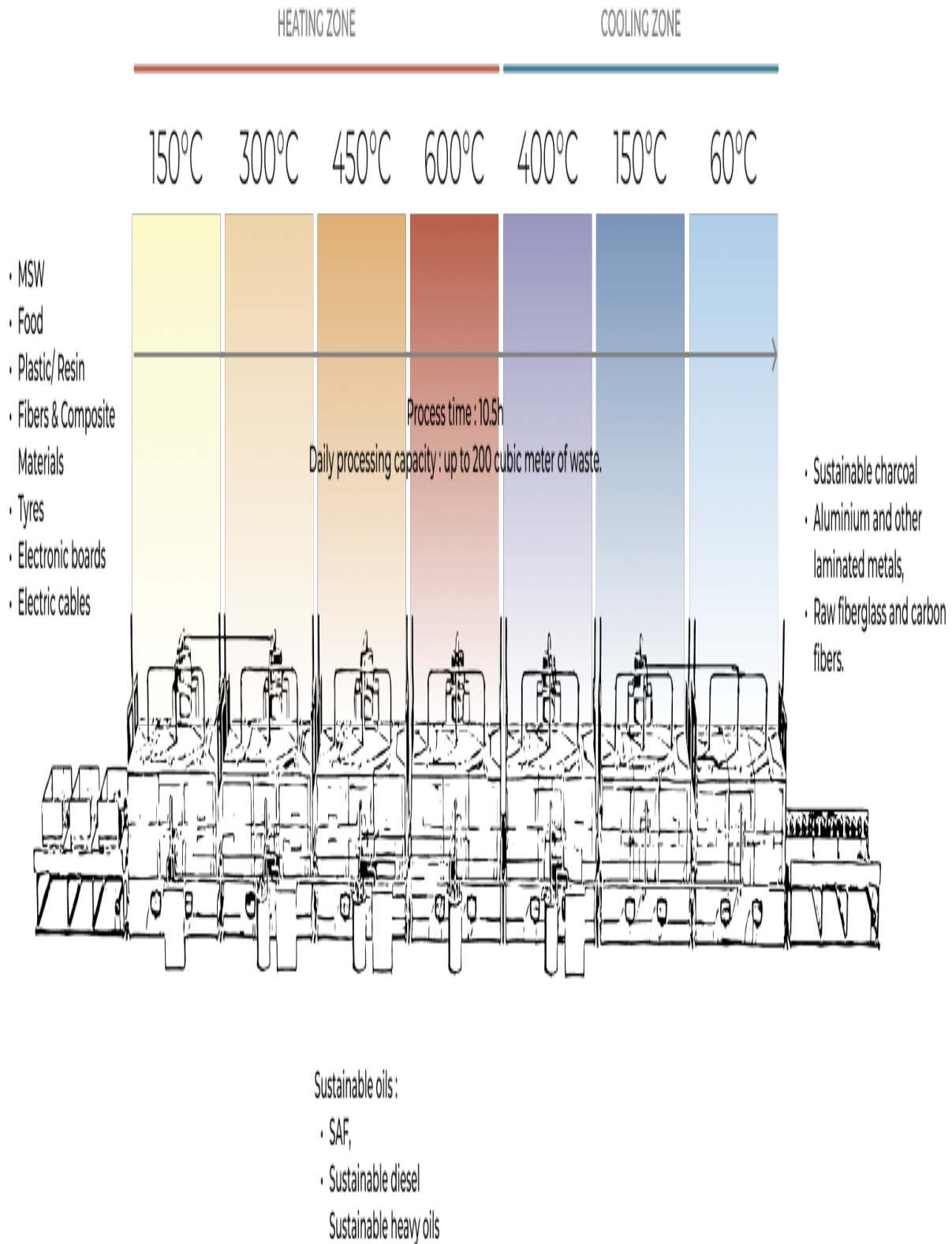


Figure 1: Illustration 2

The core of the system is a horizontal, multi-zone reactor through which waste is continuously conveyed. Three buckets move as one batch through seven temperature zones every 90 minutes, enabling 24-hour continuous throughput of 48 batches (17 units of 3 buckets).

Source: OneWorld Corporation technical documentation; Urban Rig product brochures.

4.2 Key Process Advantages

No Sorting Required. Mixed waste — including mud, non-resin materials, metals, pesticides, and food residue — is loaded directly into transport buckets. The only preparation is one day of storage to reduce surface moisture.

Handles Any Waste Condition. The URC-2000 is specifically engineered for carbonizing waste with high moisture content, a challenge that conventional pyrolysis systems cannot handle.

Processes PVC and All Halogenated Plastics. The catalytic layer adsorbs chlorine, while steam neutralises hydrochloric acid. Recent testing confirms this extends to all halogenated compounds — including brominated and fluorinated plastics used in fire-retardant applications.

Precision Automation, Minimal Staffing. The entire 7-process cycle is automatically operated by microcomputer control. A URC-2000 (200 T/day nominal) operates with approximately 15 staff.

Source: Urban Rig Recycling Solutions ENG; UR2000-JP technical specification; Asada Shokai test report.

4.3 Four Marketable Oil Fractions

The 7-zone continuous process separates recovered oil into four distinct fractions by temperature-controlled condensation:

Zone	Temperature	Share	Oil Type	Calorific Value	Primary Uses
Zone 1	150°C	~10%	Light naphtha (C5–C8)	44–46 MJ/kg	Petrochemical feedstock; Industrial solvent

Zone	Temperature	Share	Oil Type	Calorific Value	Primary Uses
Zone 2	300°C	~80%	Gas/diesel oil (C8-C20)	43–47 MJ/kg	Recycled diesel (NKKK certified); SAF co-processing; Industrial fuel
Zone 3	450°C	~10%	Heavy oil (C20+)	40–45 MJ/kg	Heavy fuel oil; Bitumen/asphalt feedstock
Zone 4	600°C	Traces	Asphaltenes	~32 MJ/kg	Solid fuel (carbide); Biochar

Oil quality certified by NKKK: flashpoint 46°C, sulfur <0.2%, chlorine <100 ppm, calorific value 43.27 MJ/kg — comparable to fossil diesel at 42.6 MJ/kg.

Source: URC-2000 brochure; Asada Shokai report p.28; NKKK certified test report.

4.4 Environmental Performance

Zero Dioxin Emissions. The reactor operates in a completely anoxic state — dioxins and CO₂ are not generated. This is a fundamental process advantage, not a filtration afterthought.

Emissions Far Below Regulatory Limits:

Pollutant	Urban Rig Measured	French Incin. Limit	Factor Below
HCl	<1 ppm	10 mg/m ³	>10×
SO _x	<1 ppm	50 mg/m ³	>50×
NO _x	15 ppm	200 mg/m ³	>13×
Particulates	<0.002 g/m ³	10 mg/m ³	>5×

Source: KANSO Technos independent exhaust gas analysis report.

Optional PRIMO Exhaust Upgrade. For sensitive urban or airport environments, the PRIMO (PUREM 30.8) exhaust purification module performs DeNO_x, DeSO_x, dust and odour removal — enabling near-zero atmospheric impact.

Integrated Water Purification (SEA CORAL). Ceramic purification system enables closed-loop water management, eliminating wastewater discharge.

Low Noise and Odour. Thermal decomposition occurs entirely inside enclosed equipment. Operating noise is comparable to a cooling tower.

4.5 Safety

No Explosion Risk. No oxygen in the reactor means no risk of explosion or ignition — a fundamental safety advantage over conventional thermal treatment.

Corrosion Protection. The catalytic layer, steam neutralisation, and non-oxidising atmosphere protect equipment from all halogenated compounds, ensuring long equipment life.

4.6 All Outputs Are Marketable

Every output is a saleable resource: fuel oil (four fractions), industrial-grade charcoal (carbide at ~25 MJ/kg, comparable to coal), and recovered metals in non-oxidised condition. This contrasts with conventional incineration, which produces toxic bottom ash requiring expensive hazardous landfilling. Urban Rig generates no toxic residue.

4.7 FORCE 9 Companion Technology

For specialised applications, the FORCE 9 far-infrared catalytic reduction facility can detoxify incinerated ash and carbide containing heavy metals at approximately 400°C. Treated waste becomes usable as fertilisers, soil conditioners, and construction materials.

Source: Urban Rig Recycling Solutions ENG.

5. Product Line and Scalability

Urban Rig offers a complete product range from 1 T/day batch units to 200 T/day continuous plants:

Model	Capacity	Type	Footprint / Notes
URB-10	1 T/day	Batch (stationary)	Fits 40-foot marine container
URT-10	1 T/day	Batch (mobile)	Mounted on 10-tonne truck
URB-50	5 T/day	Batch	Small industrial site
URC-400	40 T/day	Continuous (7-stage)	2,700 m ² building
URC-1000	100 T/day	Continuous (7-stage)	Mid-scale municipal
URC-2000	200 T/day	Continuous (7-stage)	4,650 m ² building

Each processing unit (~12 tonnes) is factory mass-produced. Units are split into 7 sections, transported by truck, and assembled on-site. Manufacturing lead time: approximately 10 months (continuous) or 6 months (batch). Multiple URC-2000 plants can be combined — 5× URC-2000 yields 1,000 T/day capacity on a 28,900 m² site.

Source: Urban Rig ENG, p.3 & p.19; UR2000-JP, p.7-10.

6. Feedstock Versatility

Urban Rig's anoxic superheated steam process profitably treats waste streams considered difficult or impossible to recycle through conventional means. This opens entire market segments that existing technologies cannot serve.

Municipal and Household Waste — General household waste (unsorted), food waste, mixed packaging and composites.

Industrial and Manufacturing Waste — Mixed polymer scrap, paint and coating waste, automotive shredder residue (ASR), fibre-reinforced plastics (FRP/GRP), multi-layer packaging (Tetra Pak, coffee pouches).

Tyres and Rubber — End-of-life tyres (vulcanised rubber cannot be melted or mechanically recycled). Urban Rig breaks down tyre waste into oil, carbon black, and steel.

Medical and Sanitary Waste — Contaminated plastics (less than 5% currently recycled), single-use devices, PPE, pharmaceutical blister packs. Complete thermal sanitisation at high temperature.

Aviation and Quarantine Waste — ICAO-regulated aircraft cabin waste requiring sanitisation or incineration. Urban Rig meets sanitation requirements while recovering oil and charcoal.

Agricultural Waste — Livestock waste, agricultural film, crop waste, pesticide-contaminated packaging.

Other Streams — Wire cord waste (non-oxidised copper recovery), textile waste, electronic waste plastics (brominated casings), mattresses and foam, nappies, marine and fishing waste.

Source: Urban Rig Recycling Solutions ENG; UR2000-JP; industry reports.

7. Financial Plan

7.1 Revenue Model

Urban Rig generates five distinct revenue streams from a single waste input. The following figures are based on the URC-2000 base case (200 T/day nominal, 140 T/day actual weight, 46,200 T/year at 330 operating days, mixed MSW, France):

Revenue Stream	Annual Estimate (Year 3)	% of Total
Gate fees (waste intake)	€5.3M	33%
Fuel oil sales (three fractions)	€6.3M	39%
Char/biochar sales	€1.4M	9%
Recovered metals and recyclables	€0.6M	4%
Carbon credits (CO ₂ avoided)	€2.5M	15%
TOTAL	€16.1M	100%

Revenue grows at 2% per year (inflation adjustment) from Year 2 onward. Year 1 operates at 70% capacity (ramp-up), Year 2 at 85%, full capacity from Year 3.

Source: UR-Financial-Model-Restructured-V3.xlsx.

7.2 Oil Pricing

Oil pricing is modelled at three scenarios based on current European market benchmarks (ICIS plastic pyrolysis oil index, NWE naphtha, fossil gasoil):

Scenario	Weighted Price (€/T)	Price per Litre	Annual Oil Revenue (Year 3)
Conservative (Low)	€480/T	€0.41/L	~€4.7M
Base Case (Recommended)	€641/T	€0.55/L	~€6.3M
Optimistic (ISCC premium)	€895/T	€0.76/L	~€8.8M

Under ISCC (International Sustainability and Carbon Certification), waste-derived pyrolysis oil commands a premium over fossil equivalents. As circular economy regulations tighten, this premium is expected to widen.

Source: Annex G — Oil Fraction Pricing Model; ICIS (2024–2025); ISCC framework; EU RED II/III.

7.3 Capital Requirements

Item	Amount
URC-2000 equipment (200 T/day nominal)	€50.0M
Building and site preparation	€5.0M
Installation, freight, and training	€0.9M
Accessories, spares, and monitoring	€2.0M
Pre-sorting equipment	€1.1M
Post-sorting equipment	€0.4M
Compliance and engineering (ATEX, ICPE, detailed engineering, project management)	€1.4M
Contingency (5%)	€3.0M
TOTAL CAPEX	€63.8M

7.4 Financing Structure

Component	Amount	Terms
Bank debt (70%)	€44.6M	12% annual interest, 10-year term
Equity (30%)	€19.1M	—
Annual depreciation	€4.3M	15-year straight-line

7.5 Operating Costs

Annual OPEX at full capacity (Year 3): **€7.3M**, comprising:

- Core plant staff (15 FTE): €0.87M
- Maintenance (4% of equipment CAPEX): €2.4M
- Heating oil (reactor fuel): €1.0M
- Grid electricity: €0.25M
- Insurance (0.5% of CAPEX): €0.33M

- Pre-sorting and post-sorting operations: €0.47M
- Ash and non-sellable residue disposal: €0.58M
- Contingency and other: €1.33M

Source: UR-Financial-Model-Restructured-V3.xlsx, OPEX sheet.

7.6 Energy Supply Model

The operating model uses purchased heavy fuel oil for reactor heating, ensuring reliable and consistent operations independent of feedstock composition.

- Heating demand: 2,714 MJ per tonne processed
- Annual heating fuel cost: ~€1.0M at full capacity (€0.40/L, HFO Rotterdam benchmark)
- All char/biochar sold commercially (not consumed internally): €1.4M/year at €100/T (fuel briquettes)
- All off-gas monitored for process data but not relied upon for heating

This approach maximizes product revenue while ensuring predictable operations.

Source: UR-Financial-Model-Restructured-V3.xlsx; Annex A §A.8.

7.7 Profitability and Returns

Metric	Value
Annual Revenue (Year 3, full capacity)	€16.1M
Annual OPEX	€7.3M
EBITDA	€8.9M (55% margin)
EBIT (after depreciation)	€4.6M
Net Income (Year 4, first profitable year)	€0.3M
Simple Payback (CAPEX ÷ EBITDA)	~7 years
Equity Payback	~12 years
Debt-service cash flow breakeven	Year 6
Annualised ROI on equity	17.6%

Net income turns positive in Year 4 as the initial ramp-up period completes and debt service is covered. After debt repayment ends (Year 10), free cash flow increases sharply — annual FCF exceeds €8.8M from Year 11 onward.

The 15-year cumulative free cash flow net of equity is approximately €31.3M, representing a 164% return on the initial equity investment.

Source: UR-Financial-Model-Restructured-V3.xlsx, P&L Consolidated and Cash Flow sheets.

8. Reference Deployment: CDG Airport

To illustrate the methodology, Paris-Charles de Gaulle Airport serves as a reference model. CDG exemplifies the characteristics of a strong candidate site: concentrated waste flows, high-value feedstock mix (plastics-rich cabin and packaging waste), premium gate fees (ICAO-regulated quarantine waste), proximity to industrial off-take markets, and a site operator under sustainability pressure.

8.1 Waste Availability

Charles de Gaulle Airport generates an estimated 80,000–100,000 tonnes/year of mixed waste:

- Aircraft cabin waste: 40% (32,000–40,000 T) — subject to ICAO sanitation regulations
- Terminal waste: 30% (24,000–30,000 T)
- Catering waste: 15% (12,000–15,000 T) — quarantine waste requiring thermal treatment
- Cargo and logistics: 10% (8,000–10,000 T)
- Other: 5%

A single URC-2000 (200 T/day nominal, ~140 T/day actual weight) can process the airport's entire waste stream with capacity to serve surrounding communities.

Source: Annex B estimates based on 76.2M passengers (2019), 1.2–1.5 kg waste/passenger.

8.2 Feedstock Composition (French MSW)

After full decomposition of all 13 ADEME MODECOM categories into constituent materials:

Component	% of Processable Feedstock	Primary Output
Food & Organic	38.4%	Charcoal (carbide)
Paper, Card, Wood & Cotton	37.9%	Charcoal + off-gas
Plastics & Synthetic Polymers	23.7%	Fuel oil (4 fractions)

Processable feedstock = 859.1 kg per tonne after removing 109.5 kg of glass, metals, incombustibles, and fragments, and excluding 31.4 kg mineral inert.

Source: Annex F — MSW Feedstock Yield Model (ADEME MODECOM 2017).

8.3 Yield Scenarios

Scenario	Description	Oil Yield
A: Full OMR (with decomposition)	Mixed MSW as received	23%
B: Plastic-like fraction	Pre-sorted plastic-rich stream	65%
C: PE+PP polyolefins only	High-value polyolefin stream	78%

Source: Annex F; Asada Shokai test data (9 resin types).

9. Go-to-Market Strategy

9.1 Phase 1 — Pilot Deployment (Years 1-3)

One URC-2000 unit at CDG Airport or a similar high-profile location. 200 T/day nominal capacity. Objectives: prove technology in European regulatory environment, secure ICPE permits, build stakeholder relationships, generate auditable operational data.

Target partners: ADP (Paris Airports), ADEME (grant funding), local waste authority.

9.2 Phase 2 — Regional Expansion (Years 3-7)

5-10 units across Île-de-France and major French cities (Lyon, Marseille, Lille, Toulouse). 1,000-2,000 T/day total capacity. Target: 5% of French incineration capacity replacement.

9.3 Phase 3 — European Scale-Up (Years 7-15)

Expansion to South Korea, UK, Southern Europe, and other target markets. Each additional URC-2000 replicates the proven operational and financial model. The configurable financial model adapts to local feedstock, pricing, and regulatory conditions.

9.4 Engagement Process

Discovery (2-4 weeks) → Joint scenario development (4-8 weeks) → Independent technical validation (2-4 months) → Commercial structuring and financial close → Equipment manufacture and commissioning (12-18 months to operations).

10. Risk Analysis

10.1 Technology Risk

Mitigation: Industrial-scale testing completed (Asada Shokai, 2022). Independent certification of oil quality (NKKK) and emissions (KANSO Technos). Operational plant in China. The URC-2000 is manufactured equipment, not a prototype.

10.2 Regulatory Risk

Mitigation: Urban Rig emissions are 5–50× below French incineration limits. Classification as pyrolysis (not incineration) may allow more favourable permitting. ICPE dossier preparation and DREAL consultation built into project timeline and CAPEX (€150K allocated).

10.3 Market and Pricing Risk

Mitigation: Revenue is diversified across five streams — no single stream exceeds 39% of total. Oil pricing modelled at three scenarios (€480–895/T). Gate fees are contractual and predictable. Carbon credit prices are conservative (€50/T CO₂ vs EU ETS trading at €50–80).

10.4 Feedstock Supply Risk

Mitigation: MSW volumes are large, stable, and municipally contracted. Île-de-France generates 17,000 T/day — one URC-2000 requires only 140 T/day (0.8% of regional supply). Multiple feedstock segments provide diversification.

10.5 Operational Risk

Mitigation: Automated 7-stage process reduces human error. Minimal staffing requirement (~15 FTE). 330 operating days/year assumption includes 35 days planned maintenance. Contingency provisions built into both CAPEX (5%) and OPEX (2% of CAPEX annually).

11. Appendices

The following supporting annexes provide detailed technical, financial, and market evidence:

- **Annex 1** — Process and Technology (detailed technical description)
- **Annex 2** — Technical Evidence (test data, lab analyses, emission certifications)
- **Annex 3** — Output Markets and Pricing (oil, biochar, metals, carbon credits)
- **Annex 4** — Country Studies (France and South Korea PESTEL analysis)
- **Annex 5** — Sector: Municipal Solid Waste (France)
- **Annex 6** — Sector: CDG Airport Waste (France)
- **Annex 7** — Sector: Industrial Residues (France)
- **Annex 8** — Sector: Plastic Sorting Rejects (France)
- **Annex 9** — Equipment Configuration
- **Annexes 10-13** — Scenario Economics (MSW, CDG, Industrial, Plastic Rejects)
- **Annex F** — MSW Feedstock Yield Model
- **Annex G** — CDG Airport Feedstock Model / Oil Fraction Pricing Model
- **Annex H** — Industrial Waste Feedstock Model
- **Annex I** — Plastic-Only Input Streams
- **Financial Model** — UR-Financial-Model-Restructured-V3.xlsx (15-year projection, fully configurable)